

Category-Guided Causal-Coding Transformers: Using Linguistic–Cognitive Universal Mappings for Learning, Routing, Pruning, and Continual Adaptation

Ben Goertzel

Draft, May 20, 2026

Abstract

This paper develops an architectural and training framework for causal-coding Transformer neural networks based on a categorical mapping from linguistic universals to cognitive universals. The guiding idea is that robust linguistic universals are weak but highly informative observable traces of stable cognitive invariants. If one can learn an approximate cognition–frame–language correspondence inside a Transformer, then linguistic universals can be used as training signals for latent causal modules: they can help decide which heads and channels should be stable, which dependencies should route through mediators, which context updates should commute, which couplings are likely to be double-counting, and which parts of the model should be pruned or protected.

Formally, we model a causal-coding Transformer as an object of a quantale-enriched category $\mathbf{CT}_Q$. Its modules are attention heads, value channels, feed-forward channels, adapters, predictive-coding state nodes, or column-like blocks; its context-conditioned training steps are endomorphisms $U_c : Z \rightarrow Z$; and its causal gates are based on interventional influence. We then introduce learned maps

$$\widehat{K}_\theta : \mathbf{CT}_Q \rightarrow \mathbf{Frame}_Q, \quad \widehat{E}_\theta : \mathbf{Frame}_Q \rightarrow \mathbf{Lang}_Q, \quad \widehat{\Lambda}_\theta = \widehat{E}_\theta \circ \widehat{K}_\theta,$$

where $\mathbf{Frame}_Q$ is a category of semantic frames and $\mathbf{Lang}_Q$ is a category of TUG-style linguistic cores. A linguistic universal is represented as a subobject $U \hookrightarrow L$ in $\mathbf{Lang}_Q$; its cognitive explanation is the pullback of U along $\widehat{\Lambda}_\theta$, selected by a quantale-valued weakness criterion that rewards fit and retention while penalizing cost, commutator defect, mediator leakage, and double-counting.

The main engineering thesis is this:

Use $\widehat{\Lambda}_\theta = \widehat{E}_\theta \circ \widehat{K}_\theta$ to regularize hidden Transformer updates so that causal modules, semantic frames, and linguistic externalizations form approximately commuting diagrams.

Equivalently:

Tune gates, pruning, routing, mediator heads, stability, and anti-forgetting using quantale weakness over fit, cost, commutators, mediator leakage, and double-counting.

The proposed losses include naturality losses, closure losses for hierarchy universals, mediator residual losses for cross-module dependencies, commutator losses weighted by semantic-frame

overlap, low-frustration externalization losses for word order, stability losses mapping high-stability universals into hard-kernel modules, and anti-double-counting losses over interventional footprints. The resulting framework turns category theory into a concrete recipe for causal-coding Transformer design.

Contents

1	Introduction	3
1.1	Why linguistic universals should help neural network design	3
1.2	From a philosophical correspondence to a neural objective	4
1.3	What this paper contributes	4
1.4	The intended reader	5
2	Background	5
2.1	Causal coding in Transformers	5
2.2	The Causal-Continual-Learning principle	6
2.3	Linguistic universals as structural probes	6
2.4	Quantale weakness as a unifying tuning principle	7
3	The learned cognition–frame–language diagram	7
3.1	The three categories	7
3.2	The learned correspondence	8
3.3	Naturality as the core training principle	9
3.4	Universals as subobjects and pullbacks	9
4	How each type of linguistic universal becomes a Transformer design constraint	10
4.1	Hierarchy universals become closure losses and hard-kernel structure	10
4.2	Broad cross-module universals become mediator-head constraints	10
4.3	Narrow word-order universals become low-frustration externalization losses	11
4.4	Graded universalhood becomes module stability scheduling	11
4.5	Context-indexed local laws become support-router constraints	12
5	Architecture: the Category-Guided Causal-Coding Transformer	12
5.1	High-level architecture	12
5.2	Module families	12
5.3	Causal gates with frame and linguistic influence	13
5.4	Frame and universal probes	13
5.5	Adjunction between interpretation and externalization	13
6	Losses and training objectives	14
6.1	The composite objective	14
6.2	Predictive-coding and causal-coding losses	14
6.3	Naturality loss	14
6.4	Closure loss	15
6.5	Mediator loss	15
6.6	Commutator loss	15
6.7	Double-counting loss	16
6.8	Stability loss	16
6.9	Quantale weakness objective	16

7	Theorems and design consequences	16
7.1	Naturality transports commutators	16
7.2	Closure regularization yields closed direct images	17
7.3	Mediator residuals justify pruning direct couplings	17
7.4	Category-guided commutator control implies bounded forgetting	18
7.5	Weakness selects causal mediators over double-counting	18
8	Training procedure	19
8.1	Data sources	19
8.2	A staged training recipe	19
8.3	Algorithmic sketch	20
9	Experimental program	20
9.1	A minimal experiment	20
9.2	Metrics	20
9.3	Intervention tests	21
10	Implementation choices and variants	21
10.1	Which modules should be used?	21
10.2	How symbolic must the frame layer be?	22
10.3	When typological labels are missing	22
10.4	Quantale tuning in practice	22
11	Risks, limitations, and non-goals	22
11.1	Do not over-impose universals	22
11.2	Do not confuse probes with causes	23
11.3	Do not make the frame bottleneck too narrow	23
11.4	The category theory is a design language, not magic	23
12	Conclusion	23

1 Introduction

1.1 Why linguistic universals should help neural network design

A large Transformer trained only by next-token prediction is under strong pressure to discover many useful linguistic regularities, but it is not explicitly told which internal structures should be reusable, which should be protected, which should be mediated, and which should be pruned as redundant. A causal-coding Transformer adds one important ingredient: it tries to identify which internal units, heads, channels, or modules have genuine interventional influence, and it gates learning accordingly. However, even causal coding needs guidance about the appropriate grain of modularity.

Linguistic universals can provide exactly this kind of guidance. A robust linguistic universal is not merely a surface regularity. In the perspective developed in the companion paper on linguistic and cognitive universals, it is the observable shadow of a cognitive invariant transported through semantic frames and linguistic externalization. That is, we have a correspondence of the schematic form

cognitive causal modules $\longrightarrow$ semantic frames $\longrightarrow$ linguistic forms and typological patterns.

If this correspondence can be learned or approximated inside a Transformer, then the linguistic side gives a weak observable probe of the cognitive side. The model need not be forced to obey a hand-coded grammar. Instead, linguistic universals become soft constraints on latent causal organization.

The central engineering idea of this paper is:

Use linguistic and semantic-frame structure as weak observable probes of the latent causal modularity we want a Transformer to learn.

This gives a principled way to structure and tune causal-coding Transformers.

1.2 From a philosophical correspondence to a neural objective

The earlier category-theoretic formulation of the linguistic–cognitive correspondence can be summarized as

linguistic universals are direct images of cognitive invariants along a weakest, quantale-enriched externalization correspondence,

and conversely

cognitive universals are weakest pullback explanations of stable linguistic subobjects.

For a neural network designer, this says the following.

First, if the network contains a stable latent causal pattern, and if that pattern is relevant to language, then some image of it should be visible through semantic-frame probes or linguistic probes. For example, a stable person-role representation should help explain person hierarchy regularities; a stable event-role representation should help explain argument-structure regularities; a stable context router should help explain context-indexed local laws.

Second, if a linguistic universal is robust across languages and contexts, then the weakest adequate explanation of that universal should correspond to some latent causal structure. If the model explains a robust hierarchy by scattering many redundant correlations across unrelated heads, that explanation is not weak: it is expensive, brittle, and double-counted. A better model should place the relevant structure in a stable module or small family of modules.

Third, if a broad linguistic dependency crosses two modules, the model should not couple those modules densely unless necessary. It should try to find a mediator. This is the neural version of the linguistic theorem that nontrivial cross-module universals require mediation.

1.3 What this paper contributes

This paper turns those conceptual points into a concrete causal-coding Transformer design. The main contributions are:

1. A quantale-enriched category $\mathbf{CT}_Q$ of causal-coding Transformers, whose objects contain modular latent state, context update endomorphisms, do-influence gates, probes, and stability assignments.

2. A learned cognition–frame–language map

$$\widehat{\Lambda}_\theta = \widehat{\mathcal{E}}_\theta \circ \widehat{\mathcal{K}}_\theta : \text{CT}_Q \rightarrow \text{Lang}_Q,$$

where $\widehat{\mathcal{K}}_\theta$ reads semantic frames from Transformer modules and $\widehat{\mathcal{E}}_\theta$ predicts linguistic externalizations from frames.

3. A set of training losses derived from categorical structure: naturality, pullback, closure, mediator residual, commutator, double-counting, stability, low-frustration externalization, and context gluing.
4. A theorem package showing, under explicit assumptions, that these losses push the model toward small context commutators, mediated cross-module dependencies, closed hierarchy representations, and lower double-counting.
5. A practical training and evaluation protocol for testing whether category-guided causal coding improves continual learning, multilingual transfer, pruning, head specialization, and interpretability.

1.4 The intended reader

The paper is written for two audiences. Readers interested in architecture can treat the category theory as a precise design language for heads, gates, probes, routers, and regularizers. Readers interested in mathematics can view the architecture as an approximate neural realization of a lax monoidal, quantale-enriched correspondence between cognitive causal systems, semantic frames, and TUG-style linguistic cores.

Every formal definition is accompanied by a plain-language explanation. The goal is not to make a Transformer into a symbolic grammar engine. The goal is to use the stable parts of linguistic structure as weak constraints on the model’s latent causal organization.

2 Background

2.1 Causal coding in Transformers

A standard Transformer contains a rich supply of potential modules: attention heads, value channels, key-query subspaces, feed-forward neurons, low-rank adapters, MoE experts, residual stream subspaces, and layer-wise prediction-error nodes in predictive-coding variants. Causal coding treats these as candidate causal factors and asks which ones have interventional influence on predictions, frames, or downstream internal states.

Let z_m denote the activation of module m . A do-influence score from m to an output or internal target r can be written schematically as

$$C_{m \rightarrow r}(x) = \left\| \frac{\partial}{\partial \alpha} r(\text{do}(z_m \leftarrow z_m + \alpha v_m); x) \Big|_{\alpha=0} \right\|,$$

where v_m is a perturbation direction. In practice this may be approximated by Jacobian-vector products, causal scrubbing, ablation, activation patching, natural-gradient surrogates, predictive-coding Hessian blocks, or randomized interventions.

Causal coding uses these influence estimates to gate updates:

$$\Delta \theta_m = -\eta g_{m,c} \nabla_{\theta_m} \mathcal{L}_c, \quad g_{m,c} \approx \sigma(a C_{m,c} - b),$$

where c is a context and $C_{m,c}$ is the estimated causal relevance of module m in context c . The goal is to avoid updating modules that do not causally matter for the present context. This supports continual learning because context updates that act on mostly disjoint modules approximately commute.

2.2 The Causal-Continual-Learning principle

Let Θ be parameter space and let each context c define a loss $\mathcal{L}_c$ and an update vector field

$$X_c(\theta) = -\nabla \mathcal{L}_c(\theta).$$

The commutator of two update fields is

$$[X_c, X_d] = DX_d X_c - DX_c X_d.$$

If $[X_c, X_d]$ is small, then training on c and then d is close to training on d and then c , and also close to a joint update on the mixture. If all relevant commutators are small, sequential learning has bounded path-dependence and bounded forgetting.

The categorical version replaces vector fields by endomorphisms. A context update is a morphism

$$U_c : Z \rightarrow Z$$

on a modular state object

$$Z \cong Z_1 \otimes \cdots \otimes Z_M.$$

If U_c acts nontrivially only on factors in $S_c \subseteq \{1, \dots, M\}$, and U_d acts only on disjoint factors S_d , then

$$U_c \circ U_d \cong U_d \circ U_c.$$

The approximate version measures the failure of this equality by a metric or quantale-valued defect.

2.3 Linguistic universals as structural probes

The TUG account of linguistic universals distinguishes five robust structural types.

1. Hierarchy universals behave like closure systems.
2. Broad cross-module universals require mediators.
3. Narrow word-order universals behave like low-frustration externalization systems.
4. Universalhood is graded by stability rather than binary.
5. Many genuine universals are context-indexed local laws.

These are not merely linguistic facts. In the correspondence view, they are signatures of the cognitive architecture that produces and understands language. A causal-coding Transformer can therefore use them as weak labels for how its latent modules should be arranged.

2.4 Quantale weakness as a unifying tuning principle

A commutative quantale is a complete lattice $(Q, \leq)$ with a monoidal product $\otimes$ distributing over arbitrary joins. Quantales let us combine heterogeneous notions of cost, evidence, weakness, confidence, and error.

For a relation $H \subseteq X \times X$ interpreted as indistinction and a valuation $\mu : X \rightarrow Q$, define

$$W(H) = \bigvee_{(x,y) \in H} \mu(x) \otimes \mu(y).$$

Large $W(H)$ means the hypothesis leaves many valued distinctions unmade. In ordinary Occam terms, it commits to less. In neural-network terms, it avoids encoding unnecessary couplings.

For Transformer tuning we use a product quantale such as

$$Q_{total} = Q_{fit} \times Q_{mret} \times Q_{mcomm}^{op} \times Q_{mleak}^{op} \times Q_{mdc}^{op} \times Q_{mcost}^{op},$$

where fit and retention should be high, while commutator defect, mediator leakage, double-counting, and cost should be low. The superscript *op* reverses the order on cost-like coordinates. A good model is a maximal element of this product order, or an approximate Pareto point selected by a scalarization or epsilon-constraint rule.

3 The learned cognition–frame–language diagram

3.1 The three categories

We now define the categorical objects used by the architecture. The definitions are intentionally flexible: different implementations may instantiate heads, channels, adapters, or columns as modules.

Definition 3.1 (The category of causal-coding Transformers). A causal-coding Transformer object is a tuple

$$T_{heta} = (Z, \{Z_m\}_{m=1}^M, \{U_c\}_{c \in C}, \{S_c\}_{c \in C}, G, \Pi, \rho, \mu),$$

where:

- $Z \cong Z_1 \otimes \cdots \otimes Z_M$ is a modular latent state object;
- each Z_m is a module, such as a head, channel group, adapter, expert, predictive-coding node, or column;
- each $U_c : Z \rightarrow Z$ is a context-conditioned update endomorphism;
- $S_c \subseteq \{1, \dots, M\}$ is the intended support of context c ;
- $G = (g_{m,c})$ is a causal-gate system derived from do-influence estimates;
- Π is a family of probes and readouts;
- $\rho_m \in [0, 1]$ is a stability or rigidity assignment for module m ;
- μ is a quantale-valued valuation on modules, traces, or behaviors.

A morphism $F : T_{\theta} \rightarrow T_{\theta'}$ is a simulation, distillation, refinement, pruning map, or architectural embedding that preserves the modular/update structure up to a quantale-valued defect. The resulting quantale-enriched category is denoted $\mathbf{CT}_Q$.

Plainly: objects are not just trained networks, but trained networks equipped with a modular causal interpretation. Morphisms are transformations of such networks that preserve their causal learning behavior sufficiently well.

Definition 3.2 (Semantic-frame category). An object of $\mathbf{Frame}_Q$ is a structured semantic frame system

$$F = (R, A, E, C, \Gamma_F, \mu_F),$$

where R is a role set, A is an argument or participant structure, E is an event or force-dynamic structure, C is a context index set, Γ_F is a symmetry or equivalence structure, and μ_F is a quantale valuation. Morphisms are frame embeddings, refinements, role maps, event decompositions, and context restrictions, again enriched by defects in Q .

A semantic frame is a cognitive structure such as “agent acts on patient”, “speaker addresses hearer”, “object is possessed by possessor”, “event has cause and effect”, or “clause is subordinate in this discourse context.”

Definition 3.3 (Linguistic TUG category). An object of $\mathbf{Lang}_Q$ is a TUG-style linguistic core

$$L = (C_\ell, \mathcal{O}_{\text{typ}}, \perp_\ell, \Gamma_\ell, \mu_L),$$

where

$$C_\ell = \text{TUG}_{\mathcal{O}_{\text{typ}}}(G_\ell) = \text{Deriv}(G_\ell) / \equiv_{\perp, \Gamma, \mathcal{O}_{\text{typ}}}$$

for a typological observation interface

$$\mathcal{O}_{\text{typ}} = (\Sigma_{\text{LF}}, \Sigma_{\text{PF}}, \Sigma_{\text{Morph}}).$$

Morphisms in $\mathbf{Lang}_Q$ are interface-preserving maps between linguistic cores, including language-to-language alignments, quotient maps, typological-feature maps, and translation-like maps that preserve observations up to Q -defect.

This category does not treat language merely as strings. It treats language as equivalence classes of derivations under an observation interface.

3.2 The learned correspondence

The ideal correspondence is

$$\mathbf{CT}_Q \xrightarrow{\hat{\mathcal{K}}_\theta} \mathbf{Frame}_Q \xrightarrow{\hat{\mathcal{E}}_\theta} \mathbf{Lang}_Q.$$

The composite is

$$\hat{\Lambda}_\theta := \hat{\mathcal{E}}_\theta \circ \hat{\mathcal{K}}_\theta.$$

Here $\hat{\mathcal{K}}_\theta$ is a learned frame extractor. It maps hidden module states to semantic-frame predictions:

$$\hat{\mathcal{K}}_\theta(z) = \hat{F}_z.$$

The externalization map $\widehat{\mathcal{E}}_\theta$ maps semantic-frame predictions to linguistic-core predictions:

$$\widehat{\mathcal{E}}_\theta(\widehat{F}_z) = \widehat{L}_z.$$

Thus

$$\widehat{\Lambda}_\theta(z) = \widehat{L}_z.$$

In practice these maps may be implemented as auxiliary heads, linear probes, sparse adapters, contrastive encoders, structured decoders, or neural-symbolic readouts. They need not form a hard bottleneck. Their job is to make the hidden causal modules partially legible in semantic and linguistic terms.

3.3 Naturality as the core training principle

Suppose context c acts on hidden state by $U_c : Z \rightarrow Z$. Suppose the corresponding semantic or linguistic transformation is V_c . The key square is

$$\begin{array}{ccc} Z & \xrightarrow{U_c} & Z \\ \widehat{\Lambda}_\theta \downarrow & & \downarrow \widehat{\Lambda}_\theta \\ \widehat{L} & \xrightarrow{V_c} & \widehat{L}. \end{array}$$

The square commutes exactly if

$$\widehat{\Lambda}_\theta(U_c z) = V_c \widehat{\Lambda}_\theta(z).$$

In real networks it should commute approximately. Define the naturality loss

$$\mathcal{L}_{\text{nat}} = \sum_{c \in C} \mathbb{E}_{z \sim D_c} d_Q(\widehat{\Lambda}_\theta(U_c z), V_c \widehat{\Lambda}_\theta(z)).$$

Plainly: changing the hidden state according to context c and then reading out its linguistic/frame structure should approximately equal first reading out the linguistic/frame structure and then applying the corresponding linguistic/frame change.

This is the simplest way to make category theory operational in training.

3.4 Universals as subobjects and pullbacks

A linguistic universal is represented as a subobject

$$i_U : U \hookrightarrow L$$

in Lang_Q . For example, U may be the subspace of cores satisfying a number hierarchy, a person hierarchy, a case implication, a word-order coherence law, or a context-indexed local rule.

The neural preimage of U is the pullback

$$\begin{array}{ccc} \widehat{\Lambda}_\theta^{-1}(U) & \longrightarrow & U \\ \downarrow & & \downarrow i_U \\ Z & \xrightarrow{\widehat{\Lambda}_\theta} & L. \end{array}$$

This says: the hidden states that explain or instantiate the linguistic universal are exactly those whose linguistic readout lands in U .

If the universal is robust, the pullback should not be a scattered accidental region of hidden space. It should align with a stable module, support set, mediator, or context-local route. This is the core diagnostic use of the mapping.

4 How each type of linguistic universal becomes a Transformer design constraint

4.1 Hierarchy universals become closure losses and hard-kernel structure

Let P be a set of typological or frame features and let $R \subseteq P \times P$ encode hierarchy implications. If aRb , then feature a implies feature b . Let R^* be the reflexive-transitive closure and define

$$\text{cl}_R(S) = \{b \in P : \exists a \in S \ aR^*b\}.$$

A state is closed if $\text{cl}_R(S) = S$.

In a Transformer, let $p_a(z) \in [0, 1]$ be a probe's estimated probability that hidden state z carries feature a . A differentiable closure penalty is

$$\mathcal{L}_{\text{closure}} = \sum_z \sum_{aRb} [p_a(z) - p_b(z)]_+^2.$$

This says: if the model detects a marked or higher feature, it should also detect the less marked or lower feature implied by the hierarchy.

The architecture should also map high-stability closure structures into protected modules. Let $\sigma(a)$ be the empirical or modeled stability coefficient of feature a , and let r_m be the rigidity of module m . If A_{am} is an attribution weight saying feature a is represented in module m , define

$$\mathcal{L}_{\text{kernel}} = \sum_{a,m} A_{am} (\sigma(a) - r_m)^2.$$

High-stability universals are thereby encouraged to live in high-rigidity modules: hard kernels or inner shells rather than disposable outer-shell residue.

4.2 Broad cross-module universals become mediator-head constraints

Suppose a proposed dependency links module family A and module family B . The TUG factorization theorem says that if A and B are genuinely independent, no nontrivial cross-module universal should hold without a mediator. The neural design rule is:

Do not allow broad dependencies to be implemented as diffuse dense coupling.
Allocate explicit mediator heads or mediator channels instead.

Let z_A, z_B, z_M be latent states for the two endpoint modules and a mediator. The desired condition is

$$z_A \perp z_B \mid z_M.$$

A direct information-theoretic penalty is

$$\mathcal{L}_{\text{med}} = \sum_{(A,B,M)} \text{MI}(z_A; z_B \mid z_M).$$

If conditional mutual information is too expensive, one may use a surrogate based on conditional HSIC, CKA residuals, sparse regression residuals, or contrastive leakage predictors.

The categorical version uses a residual. If $A \otimes M \leq B$ is the desired implication in a quantale of behaviors, then the weakest mediator is

$$M^* = A \Rightarrow_Q B,$$

where $\Rightarrow_Q$ is the right residual:

$$A \otimes M \leq B \iff M \leq A \Rightarrow_Q B.$$

In neural terms, the model should search for the smallest mediator capacity that explains the dependency. Direct $A \leftrightarrow B$ edges are pruned unless they produce evidence gain that cannot be routed through M .

4.3 Narrow word-order universals become low-frustration externalization losses

Narrow word-order universals are neither arbitrary nor absolute. They look like local coherence pressures among externalization choices. Let $x_i \in \{-1, +1\}$ encode a word-order or externalization choice, and let $s_{ij} \in \{-1, +1\}$ encode whether choices i and j prefer to align or anti-align. A sparse signed graph energy is

$$E_{\text{ext}}(x) = \frac{1}{2} \sum_{i < j} w_{ij} (1 - s_{ij} x_i x_j).$$

A low-frustration loss is

$$\mathcal{L}_{\text{ext}} = \mathbb{E}_x E_{\text{ext}}(x) + \lambda_{\text{sparse}} \sum_{i < j} |w_{ij}|.$$

In a Transformer, the variables x_i may be read from linearization decisions, positional attention patterns, dependency-direction probes, or multilingual typological heads. The loss encourages the model to represent externalization choices as a sparse coupled system rather than as a dense entangled mass.

This also gives a design pattern: allocate specialized externalization heads whose couplings are sparse and low-frustration, and do not confuse them with high-stability semantic-frame heads.

4.4 Graded universalhood becomes module stability scheduling

A universal with high stability should be mapped to modules with slow learning rates, high consolidation, and strong protection. A low-stability local tendency can live in plastic adapters.

Let $\sigma_u \in \mathbb{R}$ be a stability coefficient for universal u , for instance

$$\sigma_u = \log \frac{\alpha_u}{\beta_u},$$

where α_u is the transition rate from violation to satisfaction and β_u is the transition rate from satisfaction to violation. Let η_m be the learning rate or plasticity of module m . A simple scheduling rule is

$$\eta_m = \eta_0 \exp(-\gamma s_m), \quad s_m = \sum_u A_{um} \sigma_u,$$

where A_{um} measures how much universal u is attributed to module m .

High-stability universals therefore reduce plasticity in the modules that represent them. This is the Transformer analogue of the HBCML hard-kernel/inner-shell distinction.

4.5 Context-indexed local laws become support-router constraints

A context-indexed universal has the form

$$X \Rightarrow Y \quad \text{inside context } c.$$

The neural counterpart is that the support router should activate the modules relevant to c , and the closure or implication loss should apply only on that support.

Let $q_c(z)$ be the router probability for context c , and let $p_X(z), p_Y(z)$ be probes. A context-local implication loss is

$$\mathcal{L}_{\text{local}} = \sum_c \mathbb{E}_z q_c(z) [p_X(z) - p_Y(z)]_+^2.$$

The router should also maintain support locality:

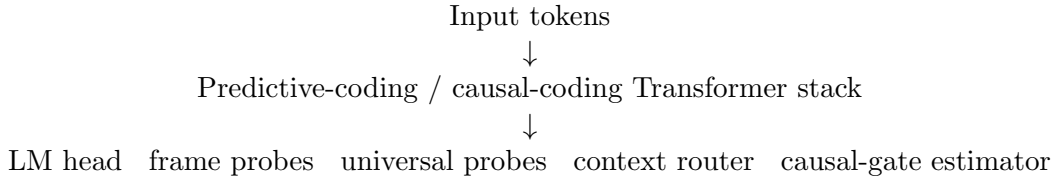
$$\mathcal{L}_{\text{support}} = \sum_c \sum_{m \notin S_c} q_c(z) g_{m,c} \|\Delta\theta_m\|^2.$$

Plainly: if a linguistic law holds only for subordinate clauses, object NPs, animate objects, or discourse-marked contexts, then the model should not smear that law across all contexts. The law should be attached to the router support that implements the relevant frame.

5 Architecture: the Category-Guided Causal-Coding Transformer

5.1 High-level architecture

The architecture can be described as follows:



Internally the Transformer stack is decomposed into modules:

$$Z \cong Z_{mkernel} \otimes Z_{mmed} \otimes Z_{mext} \otimes Z_{mshell} \otimes Z_{mtask}.$$

This decomposition is conceptual; implementation may realize it through heads, adapters, experts, low-rank subspaces, or columnar blocks.

5.2 Module families

The following module families are useful.

Hard-kernel modules. These carry high-stability frame structures: person, number, basic argument role, event role, quantification, core reference, and other closure-like invariants. They have low plasticity, strong retention regularization, and high pruning resistance.

Mediator modules. These implement bridges between otherwise factorized domains: syntax–morphology, semantics–syntax, discourse–reference, perception–language, action–language, and memory–language. They should be sparse, explicit, and monitored for leakage.

Externalization modules. These implement PF and linearization choices: word order, morphology packaging, positional conventions, local surface constraints, and low-frustration coherence.

Shell/adaptation modules. These carry medium- and low-stability patterns, including language-specific, domain-specific, speaker-specific, and task-specific residue.

Router modules. These select supports S_c for contexts, phrase types, discourse regimes, languages, domains, or tasks.

5.3 Causal gates with frame and linguistic influence

Ordinary causal coding might gate module m in context c by task influence alone:

$$g_{m,c}^{\text{task}} \propto |C_{m,c}^{\text{task}}|^p.$$

The category-guided version uses three kinds of influence:

$$C_{m,c}^{\text{task}}, \quad C_{m,c}^{\text{frame}}, \quad C_{m,c}^{\text{lang}}.$$

Define

$$g_{m,c} = \sigma \left(b_0 + b_1 |C_{m,c}^{\text{task}}|^{p_1} + b_2 |C_{m,c}^{\text{frame}}|^{p_2} + b_3 |C_{m,c}^{\text{lang}}|^{p_3} - b_4 R_{m,c} \right),$$

where $R_{m,c}$ is a redundancy or double-counting score. This gate updates a module strongly only if it causally matters for the task and for the frame/language structure it is supposed to carry.

5.4 Frame and universal probes

The model contains probes

$$\Pi_F : Z \rightarrow \hat{F}, \quad \Pi_U : Z \rightarrow [0, 1]^U,$$

where Π_F predicts semantic frames and Π_U predicts the satisfaction of a set of linguistic universals or universal-like constraints. These probes are not merely diagnostic. They participate in training through the losses described in Section 6.

The probes may be trained from:

- explicit semantic-role and frame annotations;
- dependency parses and morphological tags;
- multilingual typological labels;
- synthetic counterfactual interventions;
- self-supervised consistency across paraphrase, translation, and context restriction.

5.5 Adjunction between interpretation and externalization

Language understanding and language generation should be coupled. Let

$$\hat{\mathcal{S}}_\theta : \text{Lang}_Q \rightarrow \text{Frame}_Q$$

be an interpretation map and

$$\hat{\mathcal{E}}_\theta : \text{Frame}_Q \rightarrow \text{Lang}_Q$$

be an externalization map. The categorical ideal is an adjunction

$$\widehat{\mathcal{E}}_\theta \dashv \widehat{\mathcal{S}}_\theta.$$

This means, roughly, that producing a linguistic form from a frame and interpreting that form back into a frame are two sides of the same structured correspondence.

A practical cycle loss is

$$\mathcal{L}_{\text{adj}} = d_F(\widehat{\mathcal{S}}_\theta \widehat{\mathcal{E}}_\theta(F), F) + d_L(\widehat{\mathcal{E}}_\theta \widehat{\mathcal{S}}_\theta(L), L/\equiv_{\text{obs}}).$$

The quotient $L/\equiv_{\text{obs}}$ matters: the regenerated sentence need not be textually identical, only observationally equivalent under the selected linguistic interface.

6 Losses and training objectives

6.1 The composite objective

A category-guided causal-coding Transformer can be trained with

$$\begin{aligned} \mathcal{L} = & \mathcal{L}_{\text{LM}} + \lambda_{\text{PC}} \mathcal{L}_{\text{PC}} + \lambda_{\text{do}} \mathcal{L}_{\text{do}} + \lambda_{\text{nat}} \mathcal{L}_{\text{nat}} + \lambda_{\text{adj}} \mathcal{L}_{\text{adj}} \\ & + \lambda_{\text{closure}} \mathcal{L}_{\text{closure}} + \lambda_{\text{med}} \mathcal{L}_{\text{med}} + \lambda_{\text{comm}} \mathcal{L}_{\text{comm}} + \lambda_{\text{dc}} \mathcal{L}_{\text{dc}} \\ & + \lambda_{\text{ext}} \mathcal{L}_{\text{ext}} + \lambda_{\text{stab}} \mathcal{L}_{\text{stab}} + \lambda_{\text{local}} \mathcal{L}_{\text{local}} + \lambda_{\text{cost}} \mathcal{L}_{\text{cost}}. \end{aligned}$$

The coefficients should not be tuned blindly. They should be treated as Pareto or quantile-weight parameters, adjusted to maintain a useful balance among fit, retention, modularity, cost, and weakness.

6.2 Predictive-coding and causal-coding losses

For a predictive-coding Transformer, the free-energy or prediction-error loss has the generic form

$$\mathcal{L}_{\text{PC}} = \sum_{\ell} \|z_{\ell} - f_{\ell}(z_{\ell+1}; \theta_{\ell})\|_{\Lambda_{\ell}}^2 + \text{priors}.$$

Causal coding adds do-influence matching and sparsity. If $\widehat{C}_{ij}$ is an estimated do-influence and A_{ij} is an active edge or gate, use

$$\mathcal{L}_{\text{do}} = \sum_{i,j} (A_{ij} - \tau(\widehat{C}_{ij}))^2 + \lambda_{\text{sp}} \sum_{i,j} |A_{ij}|,$$

where τ is a thresholding or soft-gating function.

6.3 Naturality loss

As above,

$$\mathcal{L}_{\text{nat}} = \sum_c \mathbb{E}_{z \sim D_c} d_Q(\widehat{\Lambda}_{\theta}(U_c z), V_c \widehat{\Lambda}_{\theta}(z)).$$

This is the core categorical loss. It encourages hidden updates and linguistic/frame updates to agree.

6.4 Closure loss

For hierarchy implications aRb ,

$$\mathcal{L}_{\text{closure}} = \sum_z \sum_{aRb} [p_a(z) - p_b(z)]_+^2.$$

A stronger multi-step version uses the closure operator:

$$\mathcal{L}_{\text{closure}}^* = \sum_z \sum_{b \in \text{cl}_R(S_z)} (1 - p_b(z))^2,$$

where $S_z = \{a : p_a(z) > \tau\}$.

6.5 Mediator loss

The direct form is conditional mutual information:

$$\mathcal{L}_{\text{med}} = \sum_{(A,B,M)} \text{MI}(z_A; z_B \mid z_M).$$

A residual-prediction surrogate is

$$\begin{aligned} r_A &= z_A - h_A(z_M), \\ r_B &= z_B - h_B(z_M), \\ \mathcal{L}_{\text{med}}^{\text{resid}} &= \sum_{(A,B,M)} \text{HSIC}(r_A, r_B). \end{aligned}$$

This penalizes remaining dependence between A and B after the mediator has been accounted for.

6.6 Commutator loss

For context updates U_c, U_d , define

$$\delta_{cd}(\theta) = d_Z(U_d U_c \theta, U_c U_d \theta).$$

Weight this by semantic-frame and linguistic overlap:

$$\omega_{cd} = \text{Overlap}_F(c, d) \cdot \text{Overlap}_L(c, d).$$

Then

$$\mathcal{L}_{\text{comm}} = \sum_{c < d} \omega_{cd} \delta_{cd}(\theta).$$

This avoids wasting computation on contexts that are genuinely unrelated while focusing anti-forgetting pressure on confusing pairs.

6.7 Double-counting loss

Define the interventional footprint of module m as

$$\text{Foot}_m = (C_{m \rightarrow y}, C_{m \rightarrow F}, C_{m \rightarrow L}, C_{m \rightarrow Ctx}).$$

A double-counting penalty is

$$\mathcal{L}_{\text{dc}} = \sum_{m < n} \text{sim}(\text{Foot}_m, \text{Foot}_n) \text{dep}(z_m, z_n),$$

where dep may be mutual information, HSIC, CKA, correlation, or a sparse-regression predictability score.

Plainly: two heads may both be useful, but if they have the same causal footprint and mutually predictable activations, one is probably redundantly encoding the same dependency.

6.8 Stability loss

Let σ_u be a stability coefficient for universal or frame feature u , and let s_m be module stability. If A_{um} is an attribution matrix, use

$$\mathcal{L}_{\text{stab}} = \sum_{u,m} A_{um} (s_m - \sigma_u)^2 + \lambda_{\text{drift}} \sum_m s_m \|\theta_m - \theta_m^{\text{anchor}}\|^2.$$

High-stability modules are anchored more strongly. Low-stability modules remain plastic.

6.9 Quantale weakness objective

The loss terms can be summarized by a product-quantale score

$$\mathcal{Q}(T_\theta) = (\text{Fit}(T_\theta), \text{Ret}(T_\theta), \text{Conf}(T_\theta), \text{DC}(T_\theta), \text{Cost}(T_\theta))$$

with order

$$(f, r, c, d, k) \preceq (f', r', c', d', k')$$

iff

$$f \leq f', \quad r \leq r', \quad c \geq c', \quad d \geq d', \quad k \geq k'.$$

Thus better fit and retention are larger, while lower confusion, double-counting, and cost are larger in the opposite order. Training should seek approximate maximal elements of this order. Weighted sums are one practical approximation, but epsilon-constraint or Pareto-frontier methods are often more faithful.

7 Theorems and design consequences

7.1 Naturality transports commutators

Assumption 7.1 (Non-expansive readout). The learned correspondence $\hat{\Lambda}_\theta : Z \rightarrow L$ is non-expansive up to a quantale homomorphism $\varphi : Q_Z \rightarrow Q_L$:

$$d_L(\hat{\Lambda}z, \hat{\Lambda}z') \leq \varphi(d_Z(z, z')).$$

Theorem 7.2 (Commutator transport). *Let $U_c, U_d : Z \rightarrow Z$ be context updates and let $V_c, V_d : L \rightarrow L$ be their linguistic-frame images. Suppose the naturality defect for each context is at most ε_{nat} :*

$$d_L(\widehat{\Lambda}U_c z, V_c \widehat{\Lambda} z) \leq \varepsilon_{\text{nat}}.$$

Then the linguistic commutator defect is bounded by

$$\begin{aligned} d_L(V_d V_c \widehat{\Lambda} z, V_c V_d \widehat{\Lambda} z) \\ \leq \varphi(d_Z(U_d U_c z, U_c U_d z)) + 4\varepsilon_{\text{nat}}, \end{aligned}$$

up to the triangle-inequality constants of the chosen enrichment.

Proof. Insert and subtract the two latent-readout terms $\widehat{\Lambda}U_d U_c z$ and $\widehat{\Lambda}U_c U_d z$. The middle distance is bounded by non-expansiveness. The four outer terms are naturality defects for c and d . The result follows by the enriched triangle inequality. $\square$

Design consequence. If the model learns a good $\widehat{\Lambda}$, then reducing hidden commutators also reduces linguistic/frame-level path-dependence. Conversely, observed linguistic/frame noncommutation can be used to locate hidden context interference.

7.2 Closure regularization yields closed direct images

Theorem 7.3 (Approximate closure preservation). *Let R be a hierarchy relation on features P . Suppose probes $p_a(z)$ are δ -calibrated for the corresponding linguistic predicates and the closure loss satisfies*

$$[p_a(z) - p_b(z)]_+ \leq \varepsilon \quad \text{for all } aRb.$$

*Then for every chain aR^*b of length at most k ,*

$$p_a(z) \leq p_b(z) + k\varepsilon + 2\delta.$$

In particular, the direct image of the hidden representation under $\widehat{\Lambda}$ is approximately closed under R , with defect bounded linearly by path length.

Proof. Apply the one-step inequality along the chain from a to b and sum the errors. Calibration adds at most δ at the source and target predicates. $\square$

Design consequence. Hierarchy universals can be trained as Horn-style losses on probes. If calibration is good, closure at the probe level implies closure in linguistic readout.

7.3 Mediator residuals justify pruning direct couplings

Assumption 7.4 (Mediated decoder). For module families A, B and mediator M , the relevant downstream prediction Y factors as

$$p(Y \mid z_A, z_B, z_M) \approx p(Y \mid h_A(z_A, z_M), h_B(z_B, z_M), z_M),$$

and the residuals $r_A = z_A - h_A(z_M)$, $r_B = z_B - h_B(z_M)$ have small dependence.

Theorem 7.5 (Mediated pruning bound). *If $\text{MI}(r_A; r_B) \leq \varepsilon$ and the decoder is L -Lipschitz in the residual dependence channel, then pruning a direct $A \leftrightarrow B$ coupling while preserving the mediator M changes the downstream loss by at most $O(L\varepsilon)$ plus approximation error in the factorization.*

Proof. The factorized decoder depends on the endpoint modules through mediator-conditioned summaries. The only information lost by pruning the direct edge is the residual dependence between r_A and r_B . By the Lipschitz assumption and the information bound, the loss perturbation is controlled by $O(L\varepsilon)$, with an additional term for decoder-factorization error. $\square$

Design consequence. Direct broad couplings should survive pruning only when they carry residual information not explained by an explicit mediator. This operationalizes the TUG theorem that nontrivial broad universals require mediation.

7.4 Category-guided commutator control implies bounded forgetting

Let δ_{cd} be the hidden commutator defect and let ω_{cd} be the frame/linguistic overlap weight. Suppose training makes

$$\sum_{c < d} \omega_{cd} \delta_{cd} \leq \varepsilon.$$

The Causal-Continual-Learning theorem then gives a bound of the form

$$d(\Theta_{\text{seq}}, \Theta_{\text{joint}}) \leq K \sum_{c < d} \delta_{cd} + O(h^3).$$

The category-guided version refines which δ_{cd} matter: those with high ω_{cd} , i.e. contexts that overlap semantically or linguistically.

Proposition 7.6 (Overlap-targeted anti-forgetting). *If context pairs with $\omega_{cd} = 0$ have disjoint supports and pairs with $\omega_{cd} > 0$ satisfy $\delta_{cd} \leq \varepsilon_{cd}$, then sequential-vs-joint deviation is bounded by*

$$d(\Theta_{\text{seq}}, \Theta_{\text{joint}}) \leq K \sum_{c < d: \omega_{cd} > 0} \varepsilon_{cd} + O(h^3).$$

Design consequence. Linguistic/frame overlap can be used to construct the sparse confusion graph for continual learning. This is more informative than estimating interference only from raw gradient dot products.

7.5 Weakness selects causal mediators over double-counting

Definition 7.7 (Explanation score). An explanation E of a prediction or universal has score

$$W(E) = (\text{Fit}(E), \text{Ret}(E), \text{Cost}(E), \text{Conf}(E), \text{DC}(E))$$

in the product order with cost, confusion, and double-counting reversed.

Theorem 7.8 (Causal mediator beats redundant correlation). *Let E_{med} and E_{corr} explain the same observed linguistic/frame dependency with equal fit and retention. Suppose*

$$\text{Cost}(E_{\text{med}}) \leq \text{Cost}(E_{\text{corr}}), \quad \text{Conf}(E_{\text{med}}) \leq \text{Conf}(E_{\text{corr}}), \quad \text{DC}(E_{\text{med}}) < \text{DC}(E_{\text{corr}}).$$

Then E_{med} strictly dominates E_{corr} in the product-quantale order.

Proof. All coordinates are at least as good for E_{med} , and the double-counting coordinate is strictly better. Hence E_{corr} is not Pareto maximal. $\square$

Design consequence. A model that explains a dependency by one explicit mediator is preferred to a model that explains the same dependency by many overlapping heads with redundant footprints.

8 Training procedure

8.1 Data sources

A practical system can use several kinds of signal.

1. Ordinary language-modeling data for $\mathcal{L}_{\text{LM}}$.
2. Multilingual corpora, translation pairs, and paraphrase pairs for observational equivalence and adjunction/cycle losses.
3. Parsed corpora with dependency, role, morphology, or discourse labels for frame probes.
4. Typological databases or hand-curated universals for closure, stability, and low-frustration losses.
5. Synthetic counterfactuals and interventions for do-influence estimation.
6. Continual-learning streams over language, domain, genre, task, or user context for commutator and retention evaluation.

8.2 A staged training recipe

One possible staged procedure is:

Stage 0: Pretrain. Train an ordinary Transformer or predictive-coding Transformer on language modeling.

Stage 1: Instrument. Choose a modular decomposition into heads, channels, adapters, experts, or columns. Add frame probes, universal probes, context routers, and causal-gate estimators.

Stage 2: Calibrate probes. Train $\hat{\mathcal{K}}_\theta$, $\hat{\mathcal{E}}_\theta$, and $\hat{\Lambda}_\theta$ using frame labels, parses, multilingual alignments, paraphrases, and typological supervision.

Stage 3: Estimate do-footprints. Use interventions, patching, ablation, Jacobian-vector products, or predictive-coding Hessian approximations to estimate Foot_m for modules.

Stage 4: Train category-guided causal gates. Update modules using task, frame, and language influence. Penalize double-counting and prune redundant paths.

Stage 5: Add commutator training. Train across context streams while penalizing frame-weighted commutator defects.

Stage 6: Consolidate. Move high-stability structures into hard-kernel or inner-shell modules. Move task-local residue to outer-shell adapters. Prune mediator leakage and redundant footprints.

8.3 Algorithmic sketch

For each minibatch with context c :

1. Run the Transformer and collect module activations z_m .
2. Read frames $\hat{F} = \hat{\mathcal{K}}_\theta(z)$ and linguistic structure $\hat{L} = \hat{\mathcal{E}}_\theta(\hat{F})$.
3. Estimate or update do-footprints Foot_m .
4. Compute gates $g_{m,c}$ from task, frame, and language influence.
5. Compute $\mathcal{L}_{\text{LM}}, \mathcal{L}_{\text{nat}}, \mathcal{L}_{\text{closure}}, \mathcal{L}_{\text{med}}, \mathcal{L}_{\text{dc}}, \dots$
6. Update only gated modules, with stability-scaled learning rates.
7. Periodically test context-pair commutators and prune dominated couplings.

9 Experimental program

9.1 A minimal experiment

A first experiment need not be huge. Train four models on a multilingual or multi-domain corpus:

1. baseline Transformer;
2. causal-coding Transformer;
3. causal-coding Transformer with frame and universal probes;
4. full category-guided causal-coding Transformer with naturality, closure, mediator, commutator, and double-counting losses.

Use a small set of robust structures:

number hierarchy, person hierarchy, argument roles,
case/agreement, clause type, word-order relation.

9.2 Metrics

Evaluate:

- language-modeling loss;
- frame/probe accuracy;
- closure violation rate;
- forgetting under sequential adaptation;
- hidden commutator defect;
- linguistic/frame commutator defect;
- do-footprint redundancy;
- mediator leakage;
- pruning efficiency;
- transfer across languages and domains.

The sharp prediction is not merely lower perplexity. It is:

At comparable validation loss, the category-guided model should have lower commutator defect, lower double-counting, better pruning efficiency, and lower continual-learning forgetting.

9.3 Intervention tests

The strongest evidence would come from interventions.

Closure intervention. Perturb a module representing a high hierarchy feature and test whether lower hierarchy features remain coherently represented.

Mediator intervention. Ablate the mediator and test whether the cross-module dependency collapses; ablate a direct endpoint coupling and test whether the mediator preserves behavior.

Commutator intervention. Train contexts in reversed order and test whether final performance changes less than in the baseline.

Double-counting intervention. Remove one of two heads with similar do-footprints and test whether loss changes little; remove a unique-footprint head and test whether loss changes more.

Stability intervention. Fine-tune on a domain shift and test whether high-stability modules drift less while low-stability modules adapt.

10 Implementation choices and variants

10.1 Which modules should be used?

The framework does not require one specific modularization. Reasonable module choices include:

- attention heads;
- value-channel groups;
- FFN neuron groups;
- low-rank adapters;
- MoE experts;
- residual-stream subspaces;
- predictive-coding state nodes;
- HBCML-style columns or microcolumns.

A practical approach is to begin with coarse modules such as heads or adapters, then split modules whose do-footprints show internal heterogeneity and merge modules whose do-footprints are redundant.

10.2 How symbolic must the frame layer be?

The frame layer need not be fully symbolic. It may be:

- discrete and symbolic, using explicit roles and feature predicates;
- continuous, using learned frame embeddings;
- hybrid, using symbolic roles with vector-valued fillers;
- neural-symbolic, using a metagraph or typed graph as the interface.

The categorical equations require only that frame transformations and linguistic transformations be comparable through a loss. They do not require exact symbolic parsing at every step.

10.3 When typological labels are missing

If typological labels are unavailable, the system can still use proxy universals derived from data:

- cross-lingual consistency constraints;
- paraphrase equivalence classes;
- dependency parse implications;
- morphology co-occurrence hierarchies;
- learned low-frustration graphs over externalization decisions;
- context-local implication mining.

The framework is not limited to hand-coded universals. It can mine candidate universals and then use weakness selection to decide which ones are worth turning into architectural constraints.

10.4 Quantale tuning in practice

In practice, the product-quantale objective may be implemented by a controller that adjusts λ -weights. Let

$$M_i(t)$$

be monitored metrics such as fit, retention, commutator defect, double-counting, and cost. The controller updates λ_i so that marginal gains per unit cost are roughly equal:

$$\frac{\partial \text{Fit}}{\partial \lambda_i} \bigg/ \frac{\partial \text{Cost}}{\partial \lambda_i} \approx \text{constant}$$

for active constraints. When smoothness fails, use an epsilon-constraint policy: fix acceptable thresholds for critical quantities such as perplexity and forgetting, then minimize cost and double-counting subject to those thresholds.

11 Risks, limitations, and non-goals

11.1 Do not over-impose universals

The framework treats linguistic universals as weak constraints, not as hard-coded laws. Many proposed universals are false, context-indexed, or statistical. The model should learn graded stability and context scope. Hard constraints should be reserved only for very high-confidence closure structures.

11.2 Do not confuse probes with causes

A probe may detect a feature without the probed module causing that feature. This is why do-footprints matter. A module should be considered a carrier of a universal only if it has interventional influence on the relevant frame or linguistic readout.

11.3 Do not make the frame bottleneck too narrow

If $\hat{\mathcal{K}}_\theta$ is forced to capture all cognition in a small symbolic bottleneck, performance may suffer. The frame layer should be an auxiliary organizing structure, not necessarily the only channel through which language modeling proceeds.

11.4 The category theory is a design language, not magic

The categorical formulation clarifies what should commute, factor, pull back, or be mediated. It does not by itself solve optimization. The empirical question is whether the corresponding losses and architectures improve learning. The proposed experimental program is essential.

12 Conclusion

The linguistic–cognitive correspondence can be made operational for causal-coding Transformer design. The core idea is that linguistic universals are weak observable traces of cognitive invariants. If a Transformer learns an approximate map

$$\hat{\Lambda}_\theta = \hat{\mathcal{E}}_\theta \circ \hat{\mathcal{K}}_\theta$$

from hidden causal modules to semantic frames and linguistic externalizations, then linguistic universals can be pulled back into hidden space as architectural and training constraints.

The practical recipe is:

Use $\hat{\Lambda}_\theta$ to regularize hidden Transformer updates so that causal modules, semantic frames, and linguistic externalizations form approximately commuting diagrams.

And:

Tune gates, pruning, routing, mediator heads, stability, and anti-forgetting using quantale weakness over fit, cost, commutators, mediator leakage, and double-counting.

Hierarchy universals become closure losses and hard-kernel assignments. Broad cross-module universals become mediator-head constraints. Narrow word-order universals become low-frustration externalization losses. Graded universalhood becomes module-stability scheduling. Context-indexed laws become router-local constraints. Quantale weakness then chooses the weakest adequate explanation: prefer causal mediators over redundant correlations, sparse support over diffuse entanglement, and reusable kernels over brittle context-local residue.

The resulting system is not a symbolic grammar bolted onto a Transformer. It is a causal-coding Transformer whose latent modularity is shaped by the most stable observable traces of human language. If the theory is right, such models should not merely be more interpretable. They should forget less, prune better, transfer more cleanly across languages and domains, and discover more reusable internal structure.

References

- [1] A. Vaswani et al. Attention is All You Need. 2017.
- [2] J. Pearl. Causality: Models, Reasoning, and Inference. Cambridge University Press, 2009.
- [3] B. Goertzel. Causal Coding and the General Causal-Continual-Learning Theorem. Draft, 2025.
- [4] B. Goertzel, C. Derr, and Y. Taye. Hierarchical Bayesian Causal Modular Learning: A Two-Level Columnar Architecture for Continual Learning. Draft, 2025.
- [5] B. Goertzel. Formalizing Real-World Linguistic Universals. Draft, 2026.
- [6] B. Goertzel. Linguistic Universals as Quantale-Enriched Cognition–Language Correspondences: Causal Coding, TUG, and Weakest Explanations. Draft, 2026.
- [7] B. Goertzel. Type-theoretic Universal Grammar and TUG. Draft.
- [8] B. Goertzel. TyLAA: Type–Language Adjunction Archeology. Draft, 2025.
- [9] B. Goertzel. Weakness is All You Need: Notes on Quantale Weakness as a Unifying Generalized-Occam Principle for Cognitive Science and AGI. Draft.
- [10] B. Goertzel. Practical Intelligence Lives on the Pareto Frontier Between Independence Enhancement and Dependency Modeling. Draft, 2025.
- [11] B. Goertzel. The General Theory of General Intelligence: A Pragmatic Patternist Perspective. 2024.
- [12] K. Friston. The free-energy principle: a unified brain theory? Nature Reviews Neuroscience, 2010.
- [13] J. H. Greenberg. Some universals of grammar with particular reference to the order of meaningful elements. In Universals of Language, 1963.
- [14] E. Keenan and B. Comrie. Noun phrase accessibility and universal grammar. Linguistic Inquiry, 1977.
- [15] N. Chomsky. Aspects of the Theory of Syntax. MIT Press, 1965.